Control Systems Stability Analysis

Routh–Hurwitz Criterion: Step-by-Step

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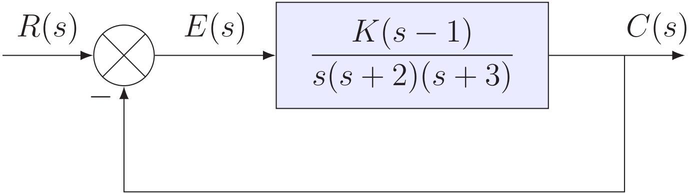
Problem Statement

Find the range of gain, K, that will cause the system to be stable.



Closed-Loop Transfer Function

Unity negative feedback:

$$T(s) = \frac{G(s)}{1+G(s)}, \qquad G(s) = \frac{K(s-1)}{s(s+2)(s+3)}$$

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Transfer function:

$$T(s) = \frac{K(s-1)}{s^3 + 5s^2 + (K+6)s - K}$$

Characteristic polynomial: $s^3 + 5s^2 + (K+6)s - K$.

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Routh Table: First Two Rows

$$s^3 + 5s^2 + (K+6)s - K$$

s^3	
s^2	
s^1	
s^0	

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s^3	1	$K+6$
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Odd-power coefficients: 1 and $K+6$.

Routh Table: First Two Rows

$$s^3 + 5s^2 + (K+6)s - K$$

s^3	1	$K+6$
s^2	5	$-K$
s^1		
s^0		

Odd-power coefficients: 1 and $K+6$.
Even-power coefficients: 5 and $-K$.

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Routh Table: The s^1 Row

s^3	1	$K+6$
s^2	5	$-K$
s^1	$-\dfrac{\begin{vmatrix} \; & \; \\ \; & \; \end{vmatrix}}{5}$	
s^0		

Routh Table: The s^1 Row

s^3	1	$K+6$
s^2	5	$-K$
s^1	$-\dfrac{\begin{vmatrix} 1 & K+6 \\ 5 & -K \end{vmatrix}}{5}$	
s^0		

Routh Table: The s^1 Row

s^3	1	$K+6$
s^2	5	$-K$
s^1	$1.2K+6$	
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$$-\frac{(1)(-K)-(5)(K+6)}{5}=\frac{6K+30}{5}=1.2K+6$$

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s^3	1	$K+6$
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s^0		

$$-\frac{(1)(-K)-(5)(K+6)}{5} = \frac{6K+30}{5} = 1.2K+6$$

Routh Table: The s^0 Row

s^3	1	$K+6$
s^2	5	$-K$
s^1	$1.2K+6$	0
s^0	$-\dfrac{\begin{vmatrix} 5 & -K \\ 1.2K+6 & 0 \end{vmatrix}}{1.2K+6}$	

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Routh Table: The s^0 Row

s^3	1	$K+6$
s^2	5	$-K$
s^1	$1.2K+6$	0
s^0	$-K$	

$$-\frac{(5)(0)-(1.2K+6)(-K)}{1.2K+6} = -\frac{K(1.2K+6)}{1.2K+6} = -K$$

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Routh–Hurwitz Criterion

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Routh–Hurwitz Criterion

The number of sign changes in the first column equals the number of poles in the right half of the s-plane. **Stability requires no sign changes**: every first-column entry must be positive.

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Stability Conditions

First-column entries 1 and 5 are already positive, so we need:

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$$\boxed{-5 < K < 0}$$